

Narendran L

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GitHub: github.com/Narendran-ds | Portfolio: narendranl-portfolio.vercel.app

Summary

Generative AI Engineer with hands-on experience building and deploying LLM-powered pipelines: a Gemini API system generating 500+ cuisines at Syncorb Geotech, and ChainSight, an explainable AI application combining a fine-tuned YOLOv8 model with Gemini-based prompt engineering for natural-language narration. Comfortable across the generative-AI workflow – dataset prep, model integration, prompt optimization, systematic testing (58 automated tests), and documentation – while working fluently with AI coding tools (Claude, ChatGPT) to accelerate development.

Skills

Languages: Python, Java, SQL

Generative AI: Gemini API, LLM Integration, Prompt Engineering, PyTorch, PaddleOCR, YOLOv8 Fine-Tuning

Data & Evaluation: pandas, NumPy, scikit-learn, XGBoost, SHAP, Dataset Preprocessing & Annotation, Model Testing/Evaluation (pytest)

AI-Assisted Development: Claude, ChatGPT (debugging, prototyping, accelerated iteration)

Backend & Deployment: Spring Boot, REST APIs, PostgreSQL, Git, GitHub, Render, Vercel, Streamlit

Frontend: React, Tailwind CSS

Experience

AI Developer Intern

Oct 2025 - May 2026

Syncorb Geotech Pvt Ltd, Chennai

- Designed and built a Gemini API pipeline to generate 500+ global cuisines and up to 500 dishes per cuisine, replacing a manual BeautifulSoup/Scrapy workflow with prompt-engineered LLM calls; added a history-tracking system so completed entries are skipped on reruns, cutting redundant API usage.
- Rebuilt the restaurant menu extraction pipeline from scratch using PaddleOCR after third-party API access was restricted, implementing folder-based batch processing and history tracking to replace an incomplete prototype with a production-ready system.
- Worked directly with pre-trained models and third-party AI APIs end-to-end: prompt design, output validation, error handling, and pipeline reliability.

Projects

ChainSight — Python, YOLOv8, NetworkX, Shapely, Gemini, Streamlit

Code: github.com/Narendran-ds/chainsight

- Architected a five-layer perception pipeline (Vision → Spatial → World Graph → Rule Engine → Narration) that converts raw warehouse video into auditable safety events, deployed live on Streamlit Cloud with 58 passing tests.
- Trained a YOLOv8 detector on a merged 17-class, 13,731-image dataset, reaching mAP@50 of 0.77 (0.89 AP@50 on exit-zone markers) after leakage-safe three-way splitting.
- Built a deterministic rule engine (R1–R5) over a NetworkX world graph with Shapely/STRtree spatial reasoning, emitting events with explicit trigger, evidence, and conclusion; constrained a Gemini Flash narration layer to rephrase-only outputs to prevent hallucinated facts.

ZipForgeX — Spring Boot, Java 17, React, PostgreSQL, NVIDIA NIM, OAuth2

Live at zipforgex.in

- Built and deployed a full-stack app (Spring Boot 3.5 backend, React + Vite + Tailwind frontend) that converts indented folder-structure text into downloadable ZIP archives, live on Render + Vercel under a custom domain.
- Integrated an AI folder-structure generator through a JWT-gated `/api/ai/generate-structure` endpoint calling NVIDIA NIM, with a system prompt locked to 4-space indentation for parser compatibility.
- Implemented JWT + Spring Security auth with Google and GitHub OAuth2, plus ZIP history and account management on PostgreSQL/Supabase, and hardened the ZIP pipeline against Zip Slip path-traversal attacks – covered by 12 passing tests.

Education

B.Tech in Artificial Intelligence and Data Science

2023 - 2027 (Expected)

Rajalakshmi Institute of Technology, Poonamallee, Chennai

CGPA: 7.44/10

Class 12

Completed March 2023

Velammal Vidyalaya, Paruthipattu

Achievements and Leadership

- Scout and Guide Leader: Led teams and demonstrated discipline in school-level camps.
- Discipline Committee Leader: Faculty-selected peer conduct manager.